

Dr Maria Korochkina

**The challenges and opportunities of recreational reading:
Insights from a 70-mln-word corpus of books popular with
British children and young people**

University of Birmingham
5 June 2026

The starting point

- Reading is a key driver of lifelong educational and economic success
- We know a great deal about how children learn to read and how they should best be taught
- Reading expertise and vocabulary growth depend on the language children are exposed to
- So... **what language do children experience when reading for fun?**

The Children and Young People's Books Lexicon

1,200 **popular** books – 400 books per age band – 70,287,217 words

7-9 years



11,162,653 words

10-12 years



21,837,794 words

13-16 years



37,286,770 words

The CYP-LEX Database

105,694 **DISTINCT** words

7-9 years



52,851 distinct words

10-12 years



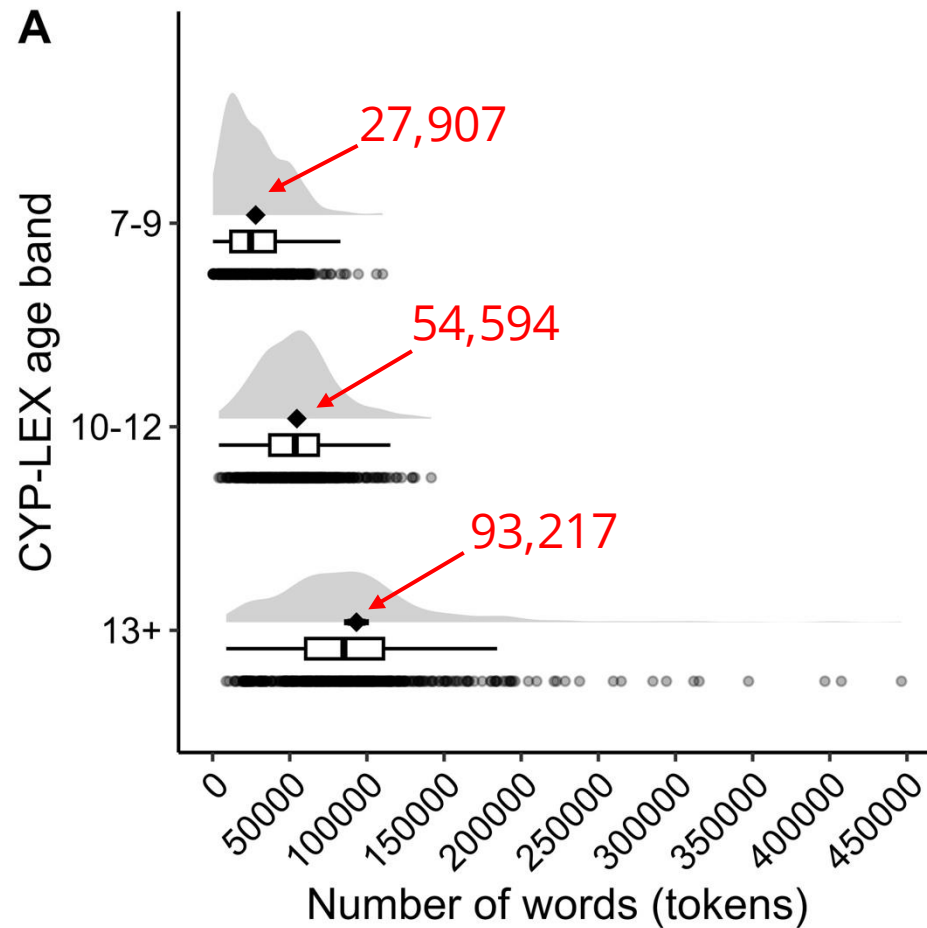
70,945 distinct words

13-16 years

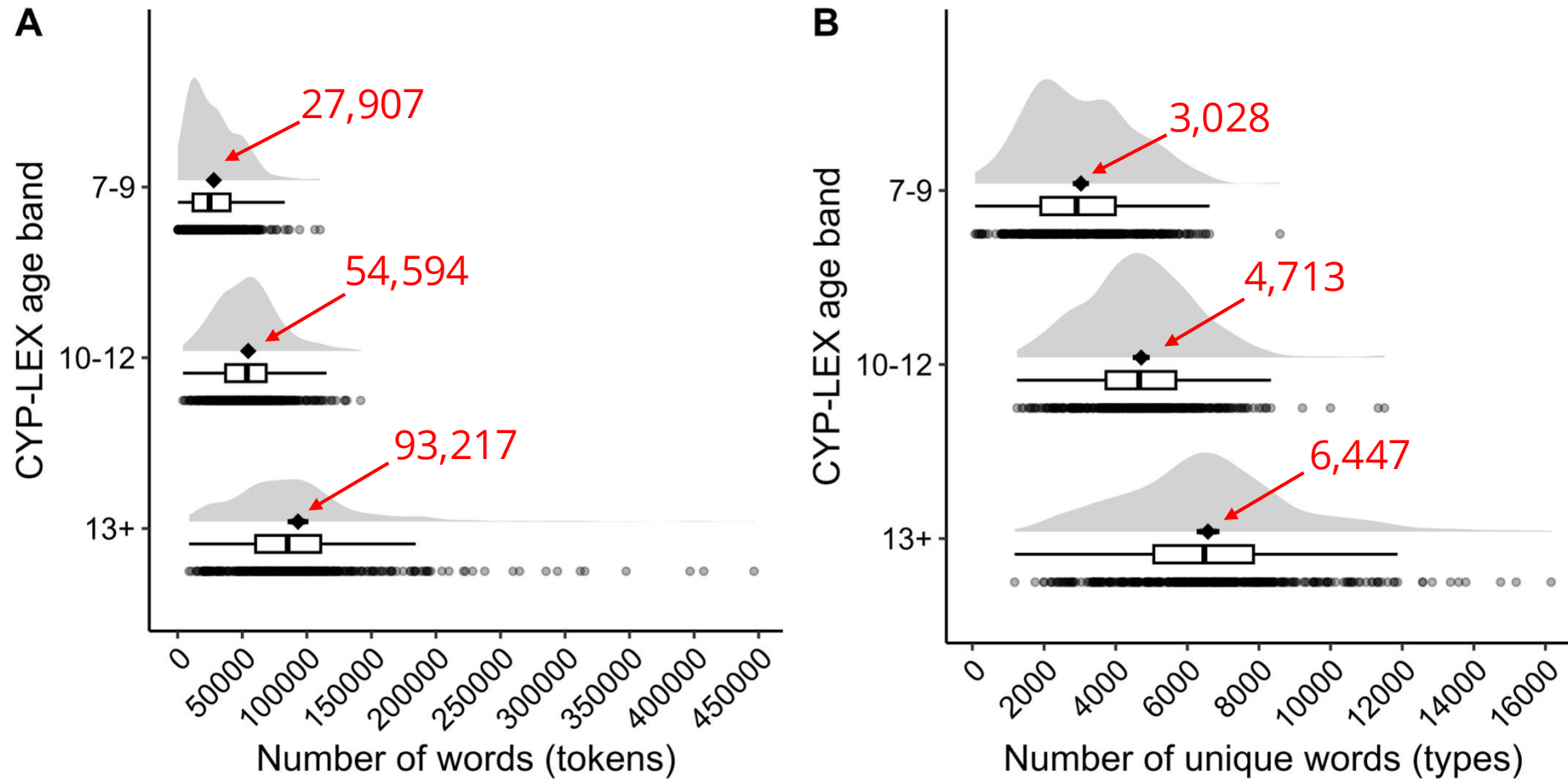


90,980 distinct words

Books contain **vast** numbers of words

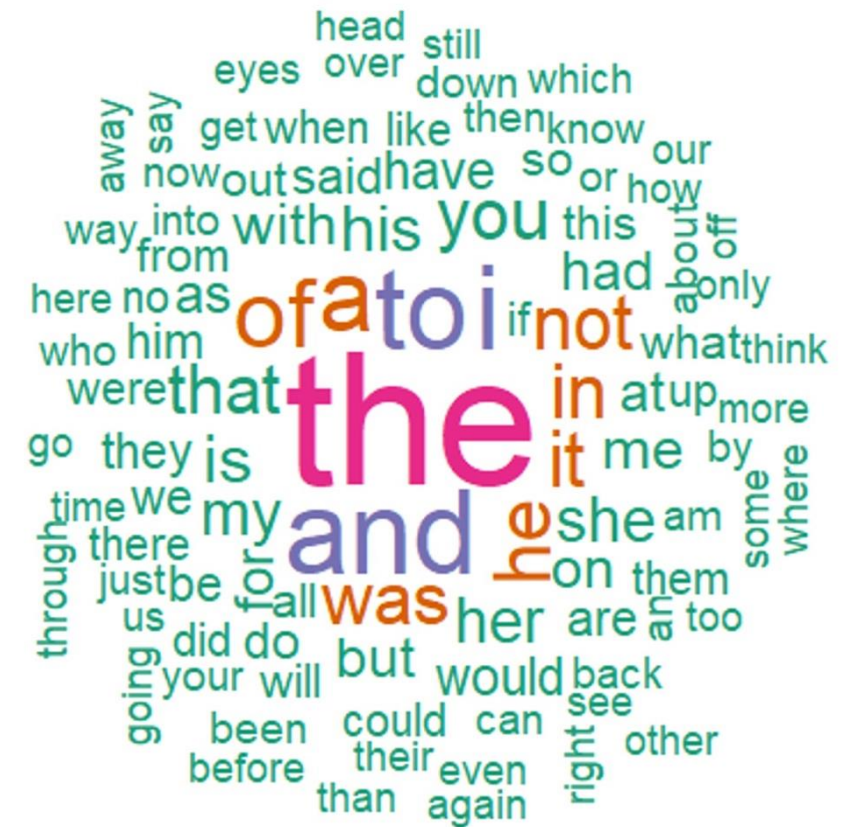


Books contain **vast** numbers of words



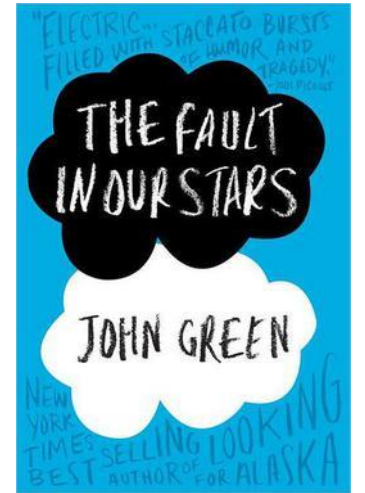
Few words are used repeatedly

- **Max 11% of words** in each age band with $fpmw > 11$
- The **most common 100 words** make up about **54%** of the text
- That's about **37 million words...**



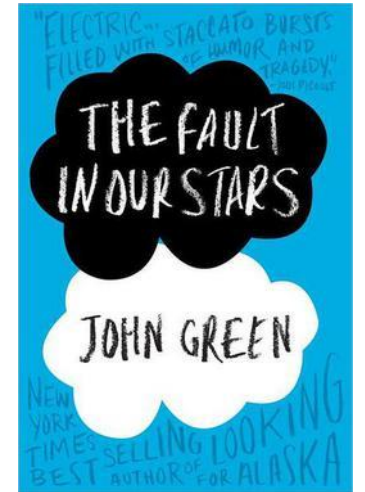
Few words are used repeatedly

“Late in the winter of my seventeenth year, my mother decided I was depressed, presumably because I rarely left the house, spent quite a lot of time in bed, read the same book over and over, ate infrequently, and devoted quite a bit of my abundant free time to thinking about death”.



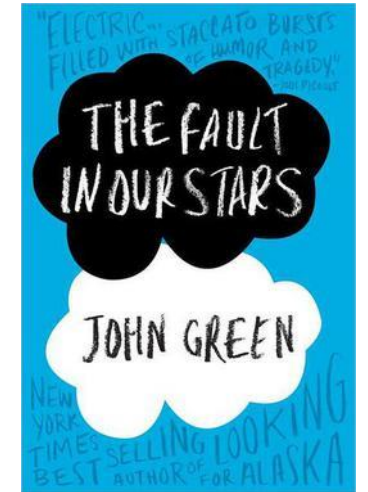
Few words are used repeatedly

" in the of my ,
my I was ,
because I the ,
a lot of in , the
same over and over, ,
and a bit of my
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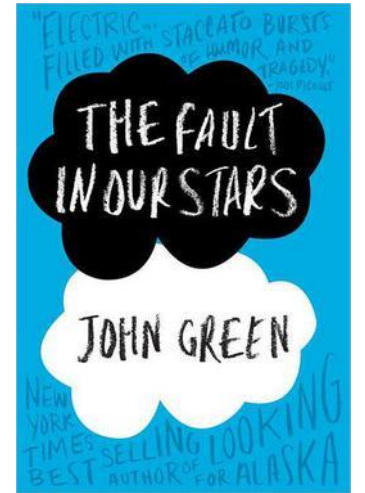


- Recognising every second word **does not take you very far**

Few words are used repeatedly

“Late in the winter of my seventeenth year, my mother decided I was depressed, presumably because I rarely left the house, spent quite a lot of time in bed, read the same book over and over, ate infrequently, and devoted quite a bit of my abundant free time to thinking about death”.

- Recognising every second word **does not take you very far**
- **Gaining repeated exposure** needed for learning **is not trivial**



Many **unfamiliar** words in books

- **40%** of distinct words don't appear on **children's TV**

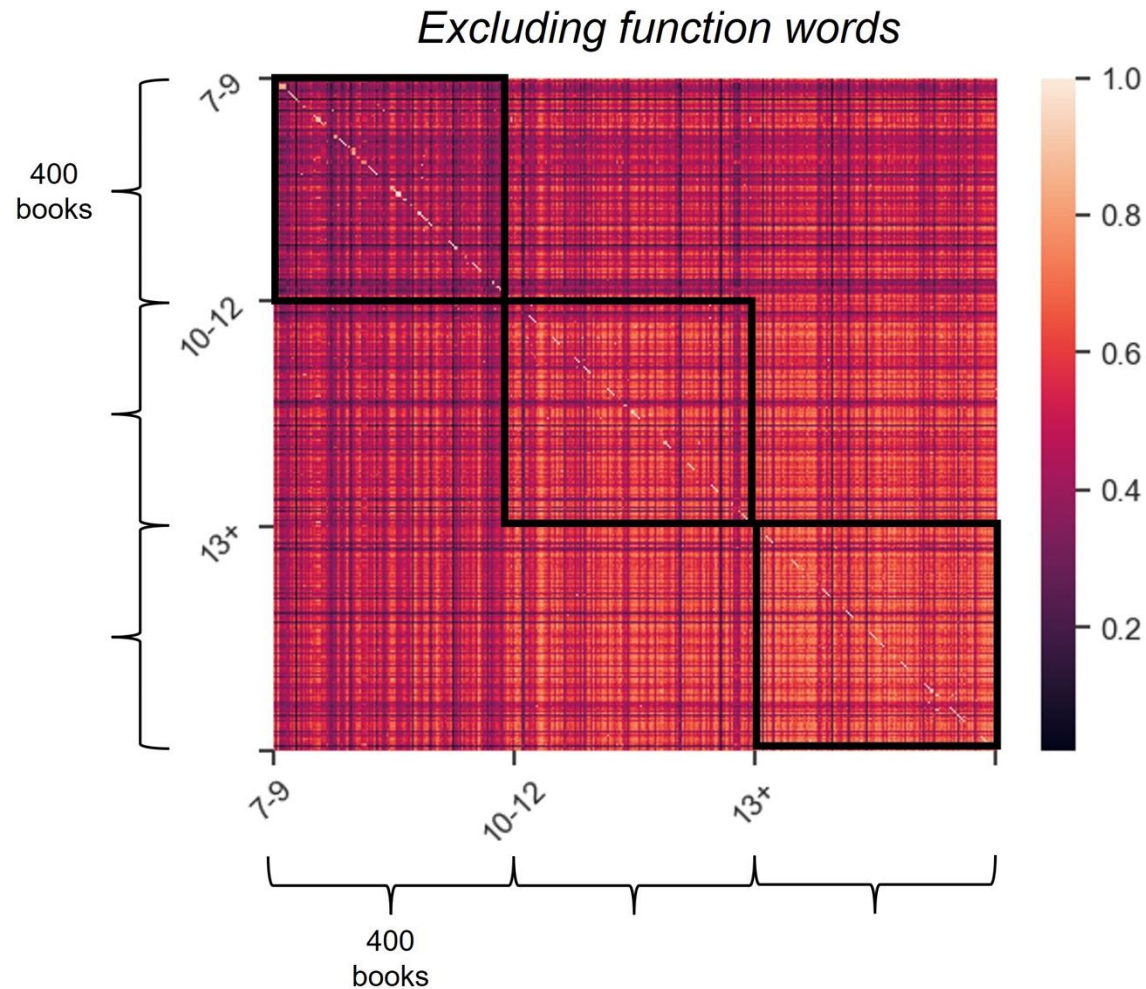


- **In 13+ books, 21%** of distinct words do not appear even in TV programmes for adults



Books offer **a unique opportunity** to build vocabulary...
... but they also present **a huge challenge** from the
earliest years of independent reading!

Books vary greatly in the words they use



- **Low similarity** in vocabulary across the individual books, **particularly in the 7-9 age band**
- In each age band, 30% of distinct words appear in **1 book out of 400** and typically **once**
- **Each book is a new challenge**

↓
To see as many words as possible as often as possible, one **must read widely**

A challenge, or a blessing?

💡 Access to **language beyond everyday use**

💡 Wonderful **opportunity to build vocabulary**

...However, the other side of the coin is that...

💡 **Strong phonics skills** are required to access even popular books

💡 Building vocabulary & reading proficiency requires **reading widely**

Questions?

Morphology in children's books



Sound + less + ly

Morphology in children's books



Sound + less + ly

Memory + use

Travel + er

Numb + ly

Guilt + y + ly

Sorrow + ful + ly

Why do morphemes matter?

- Most English words are built by **recombining morphemes**

soundlessly, tirelessly, endlessly
tasty, tasteless, tasteful

- Morpheme knowledge enables **rapid access to the meanings of familiar** words
- It is also **crucial for computing the meanings of unfamiliar** words

brightify = bright + -ify

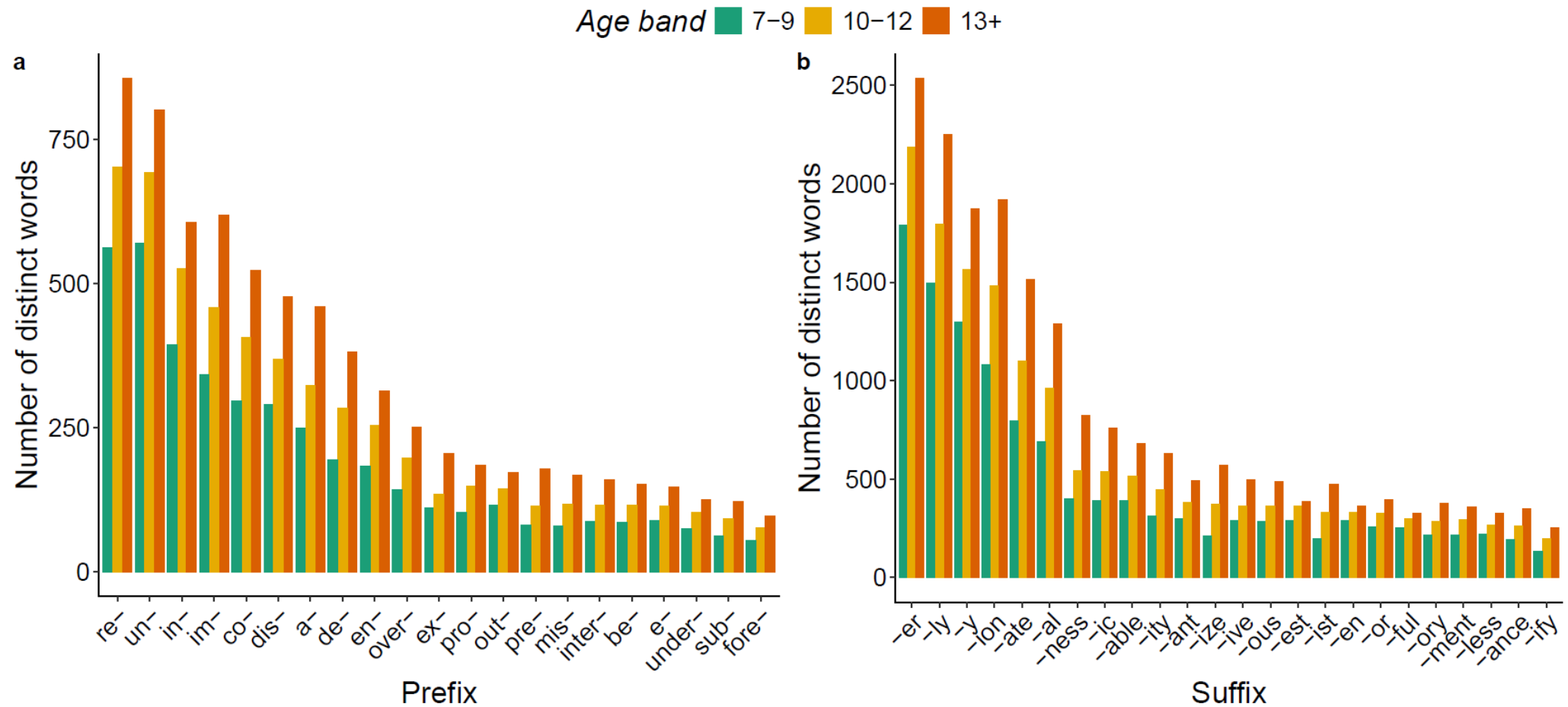
Morphology in children's books



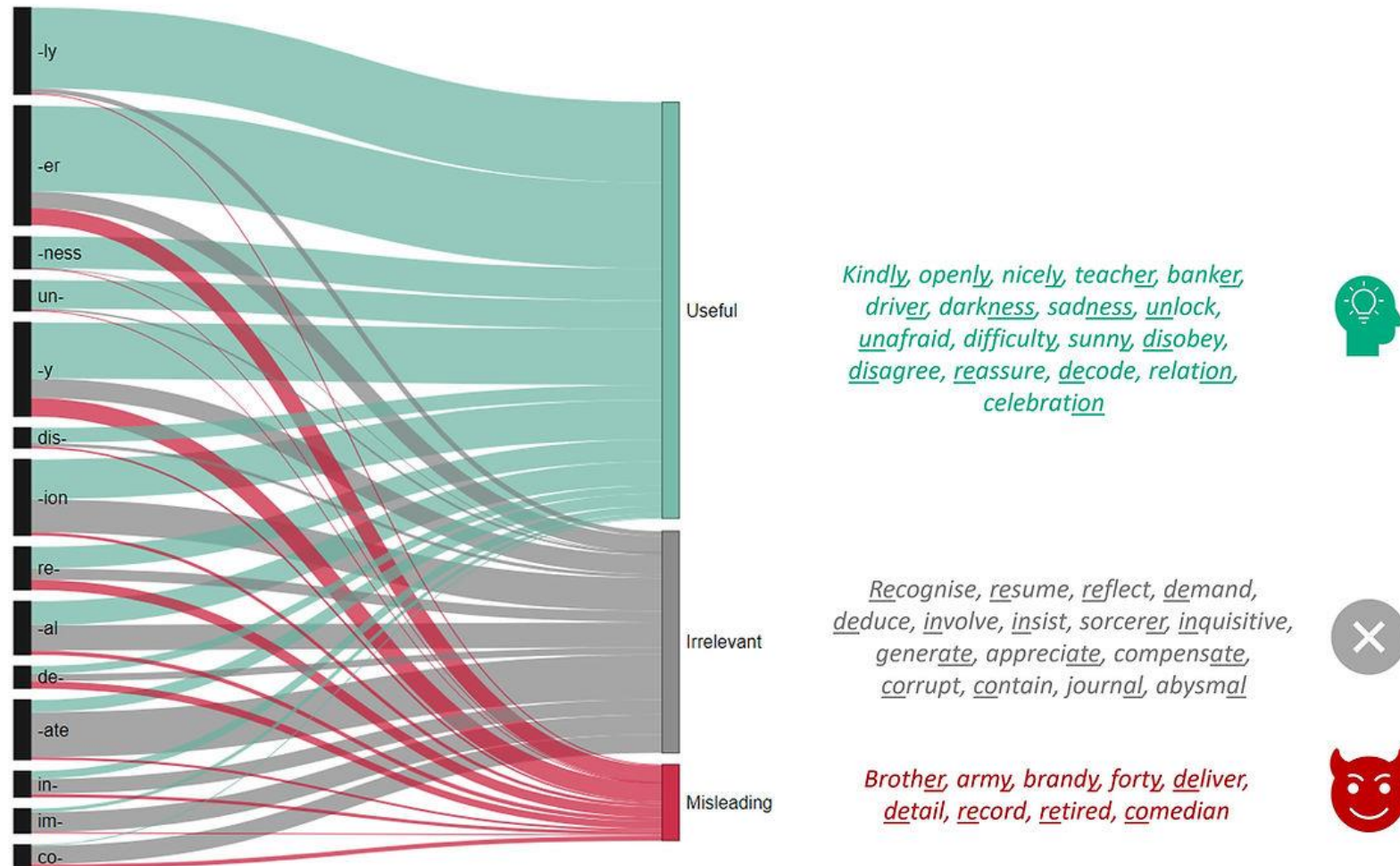
- Roughly **half** of distinct words are made up of **multiple morphemes**; most of these words **do not appear on TV**
- **Few** complex words are **used repeatedly** or in many books
- Children are **likely to see** a complex word, but **unlikely to ever see it again**

Robust **morpheme knowledge** is a must

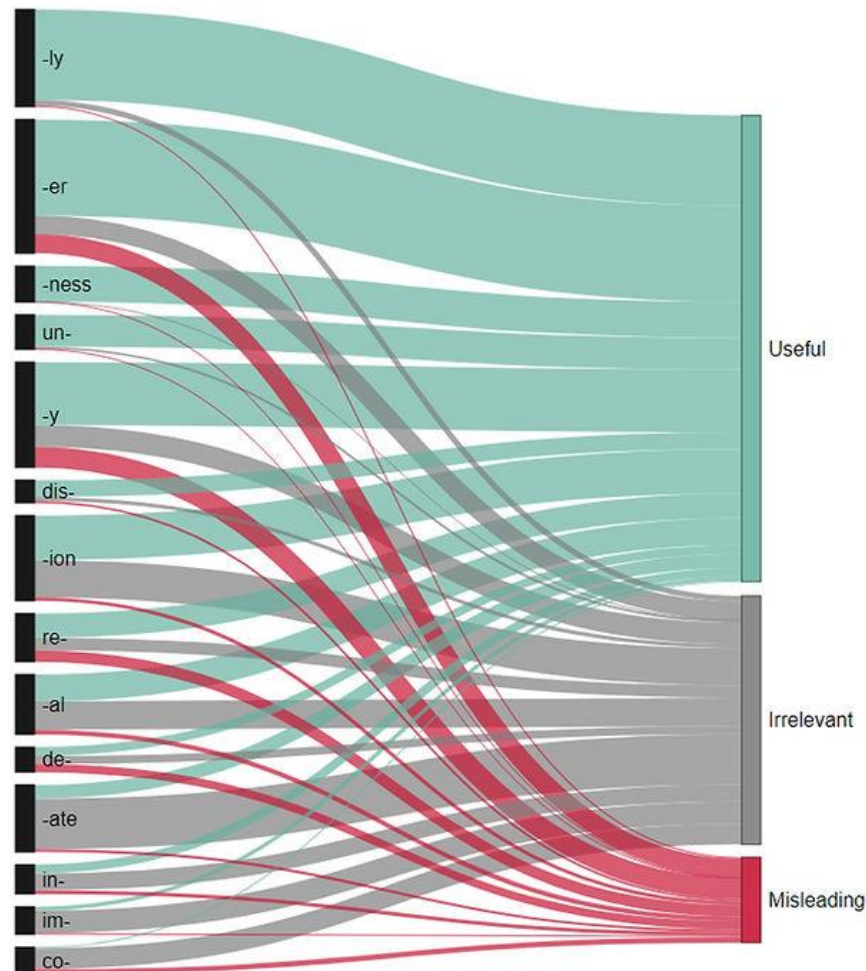
Few affixes are used frequently



Words **do not contribute equally** to learning



Some words are **useful** for learning morphemes



Kindly, openly, nicely, teacher, banker, driver, darkness, sadness, unlock, unafraid, difficulty, sunny, disobey, disagree, reassure, decode, relation, celebration



kind + **ly**
dark + **ness**

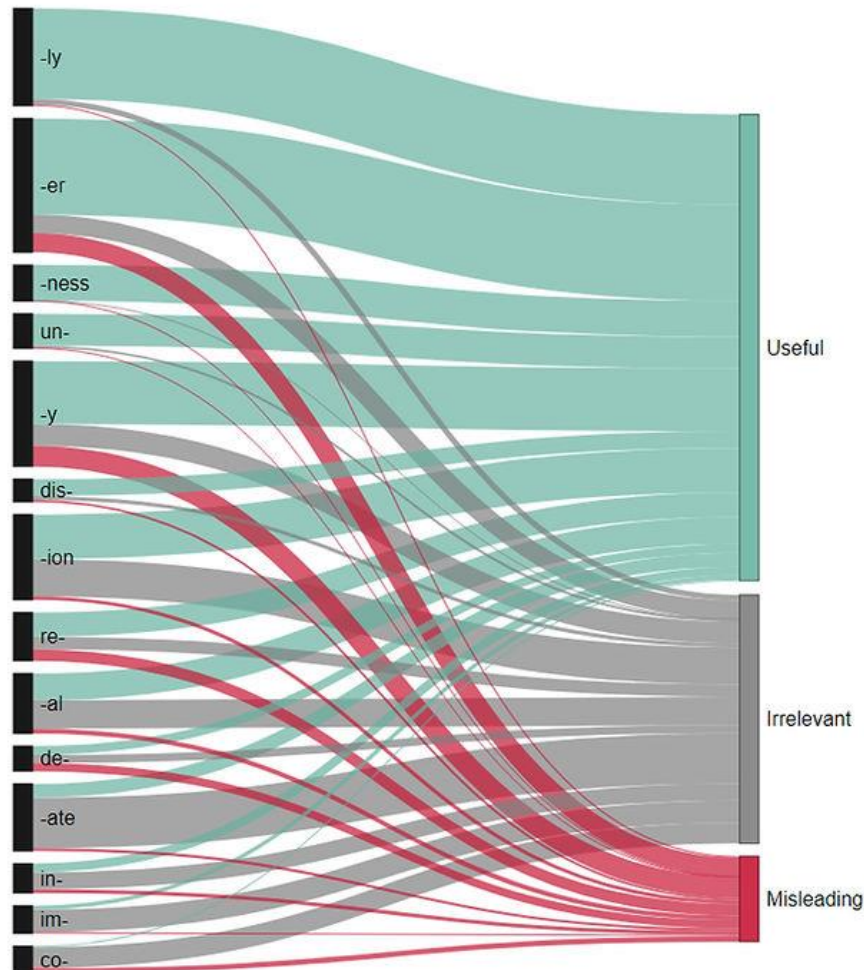
Recognise, resume, reflect, demand, deduce, involve, insist, sorcerer, inquisitive, generate, appreciate, compensate, corrupt, contain, journal, abysmal



Brother, army, brandy, forty, deliver, detail, record, retired, comedian



In many words, morphemes **can't be identified**



*Kindly, openly, nicely, teacher, banker,
driver, darkness, sadness, unlock,
unafraid, difficulty, sunny, disobey,
disagree, reassure, decode, relation,
celebration*



*Recognise, resume, reflect, demand,
deduce, involve, insist, sorcerer, inquisitive,
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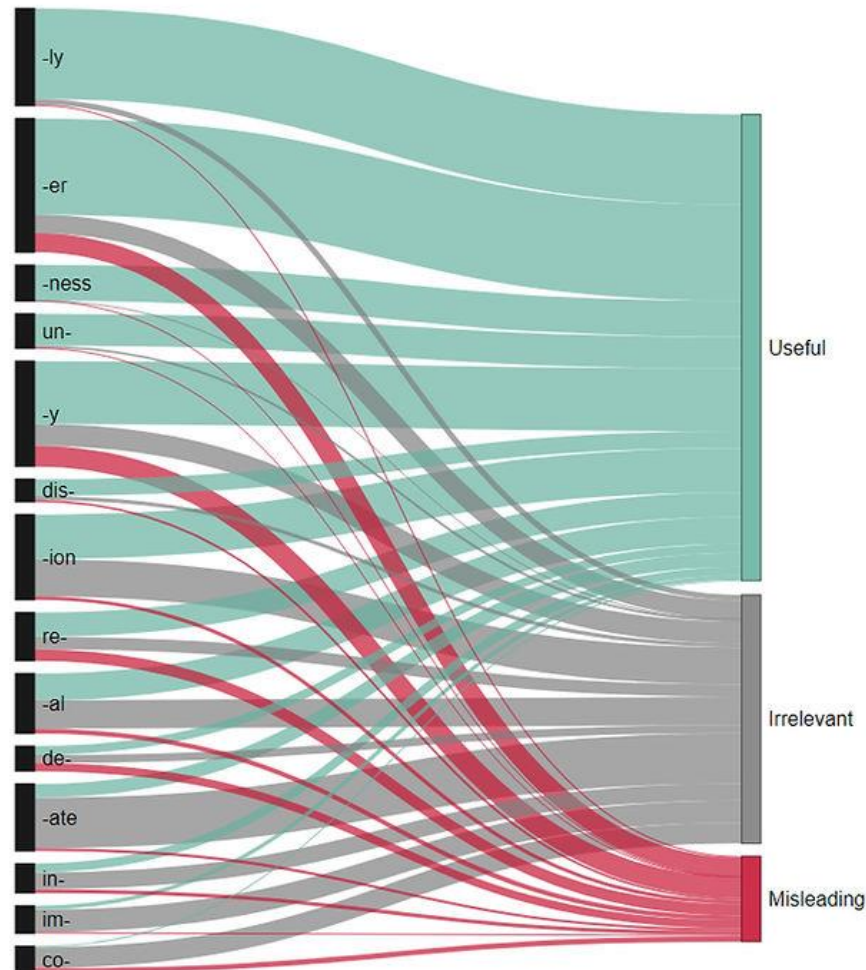


*Brother, army, brandy, forty, deliver,
detail, record, retired, comedian*



resume
demand

Some words only **appear** complex



Kindly, openly, nicely, teacher, banker, driver, darkness, sadness, unlock, unafraid, difficulty, sunny, disobey, disagree, reassure, decode, relation, celebration



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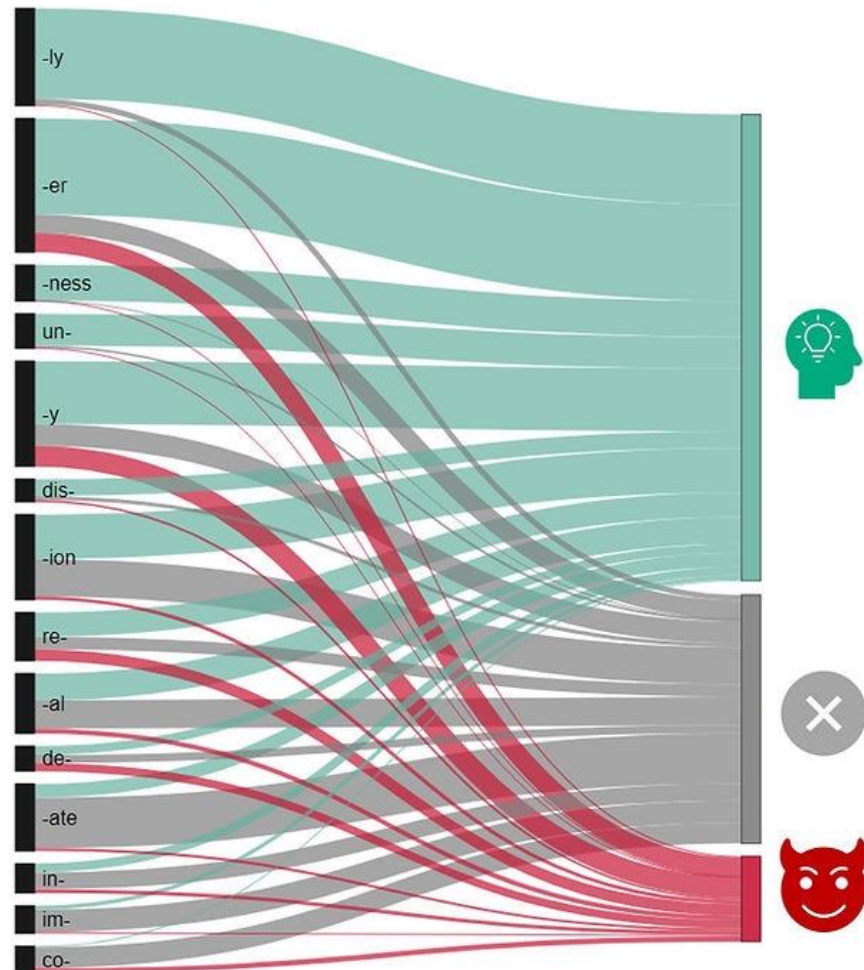


Brother, army, brandy, forty, deliver, detail, record, retired, comedian



forty
deliver

Learning morphemes is **challenging**



**Reading alone is unlikely to
be enough** to learn the
meanings of many individual
morphemes

Key insights so far...

- 💡 Access to **language beyond everyday use**
- 💡 **Wonderful opportunity** to build vocabulary, **but challenging** from the get-go
- 💡 Building vocabulary & reading proficiency requires **reading widely**
- 💡 **Strong phonics**

Key insights so far...

- 💡 Access to **language beyond everyday use**
- 💡 **Wonderful opportunity** to build vocabulary, **but challenging** from the get-go
- 💡 Building vocabulary & reading proficiency requires **reading widely**
- 💡 **Strong phonics & morphology skills** are a must
- 💡 **Plenty** of opportunities to experience **morphology**
- 💡 **Skills** to access complex words **can't be gained from reading alone**
- 💡 **Pupils will require support** to make the most of what reading can offer

Questions?

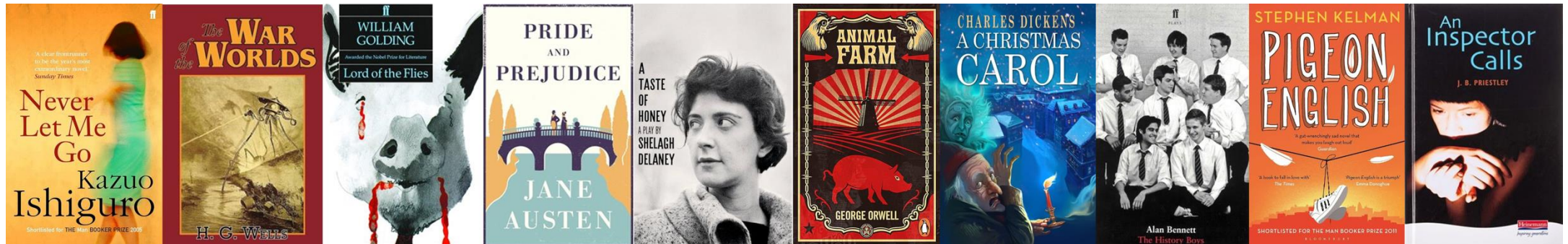
A closer look at the 13+ books

32 prose books from the AQA and EdExcel specifications
for English Literature GCSE



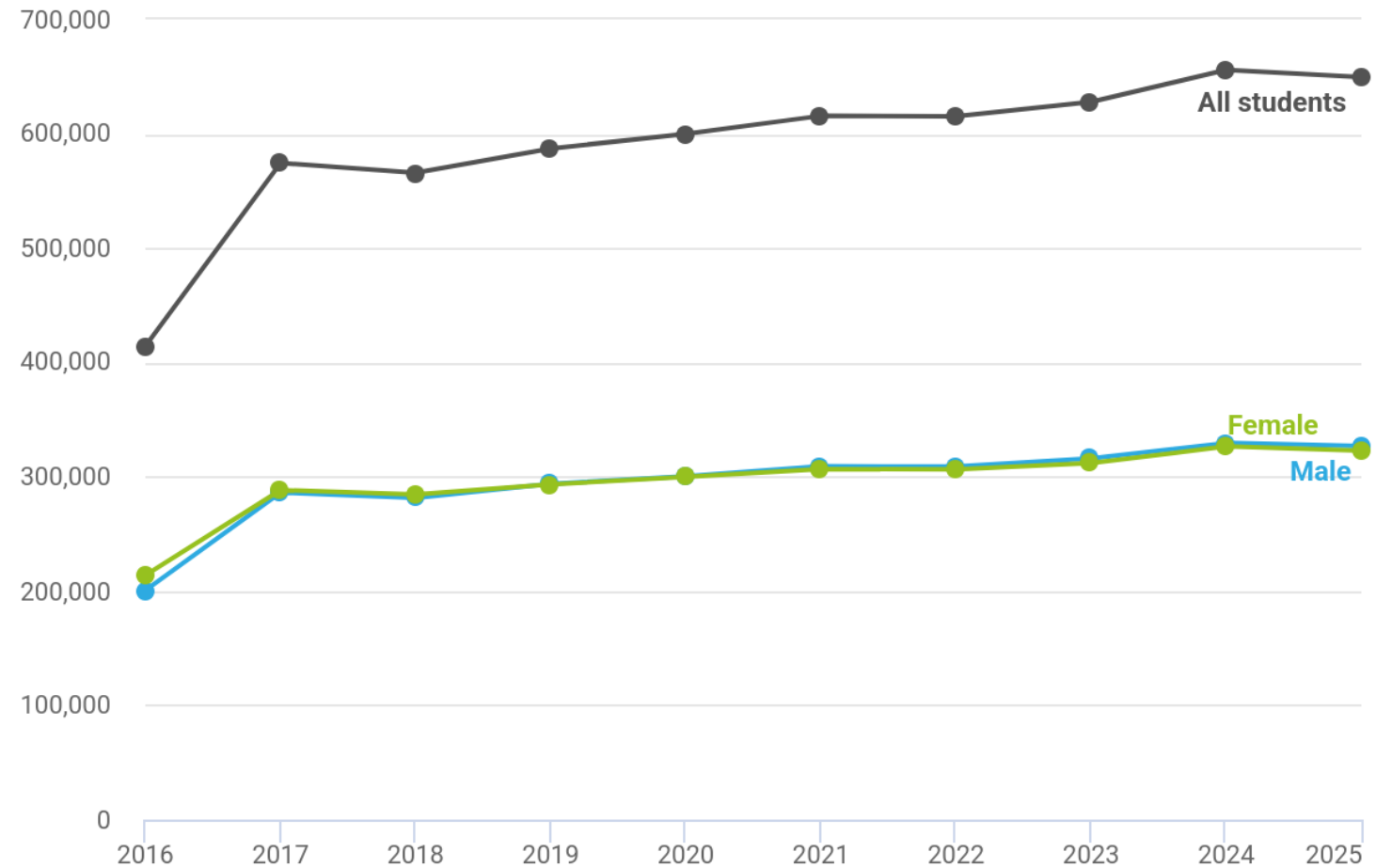
GCSE English Literature: Overview

- Curriculum set by DfE
- Qualifications delivered by exam boards (e.g., AQA, EdExcel, OCR)
- Teachers may choose from recommended texts, provided they include...
 - at least one play by Shakespeare
 - at least one 19th century novel
 - fiction or drama from the British Isles from 1914 onwards
 - a selection of post-1789 poetry (must include Romantic poetry and consist of at least 300 lines & 15 poems)



GCSE entries in English literature, 2016-2025

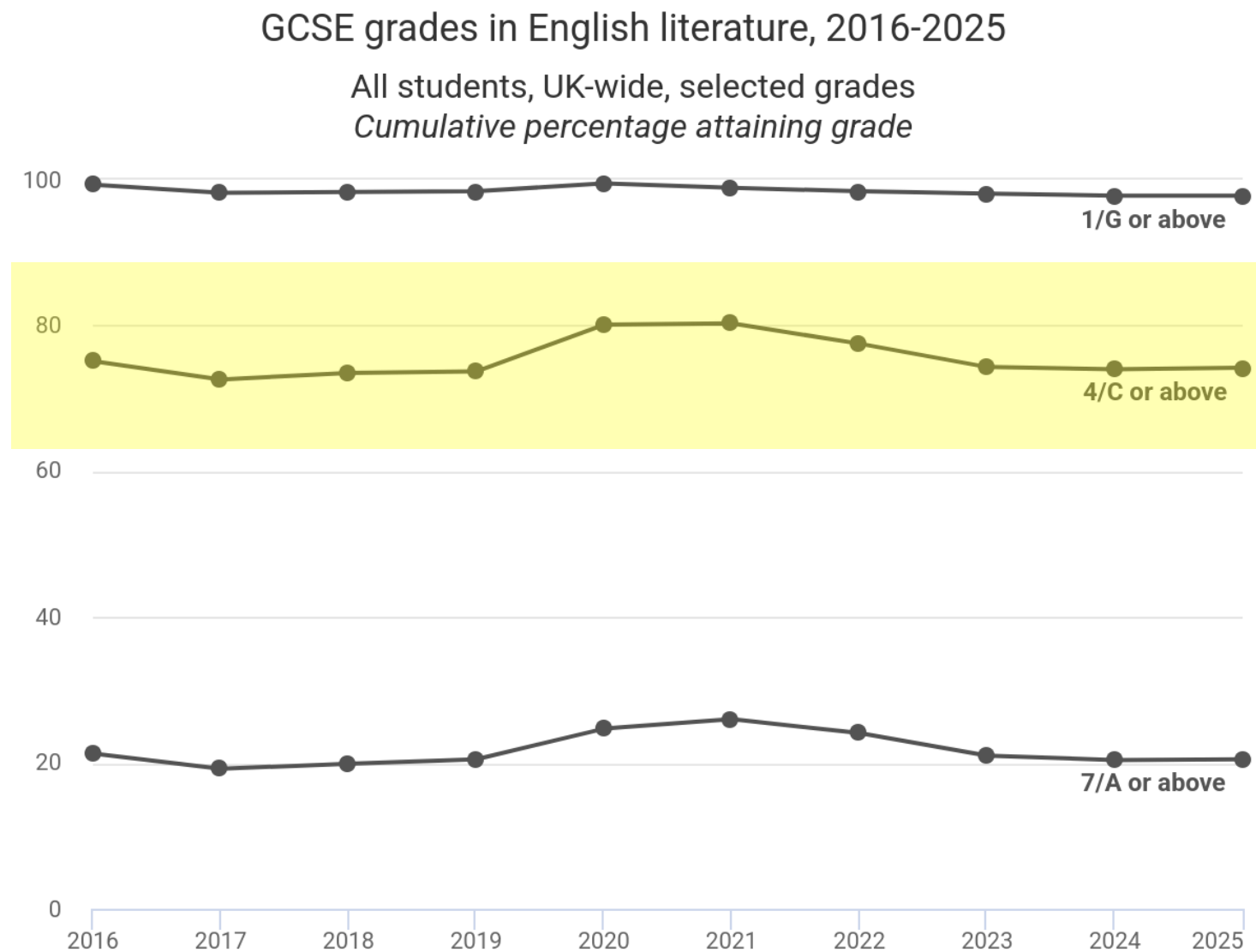
All students, UK-wide
Number of entries



**Over 500,000
entries per year
since 2017**

Source: FFT Education Datalab analysis of JCQ data
Project funded by the Nuffield Foundation

**25% of
candidates do
not achieve a
pass grade**



Source: FFT Education Datalab analysis of JCQ data
Project funded by the Nuffield Foundation

How do GCSE texts compare to popular books?

...And what makes them so challenging?

Highly dense vocabulary in the GCSE books



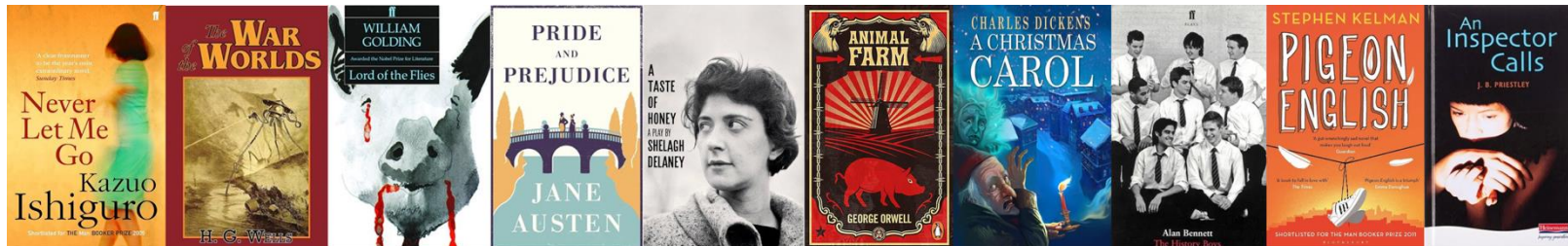
- GCSE books are **half as long as** popular books, but contain a **similar number of distinct words**
- GCSE books are much **less homogenous** in the words they use



- **More vocabulary** through **less text** in the GCSE books
- Texts **harder to understand** as a whole
- More bang for your buck **but at what cost?**

Many unfamiliar words in the GCSE books

- Only **33%** of the distinct words **occur regularly** in popular books
 - The remaining **67%** are **used sparsely**
 - **3,000** distinct words **never used on 9 BBC TV channels over 3 years**
 - *poulterer, bonneted, dowerless, bedight, sepulchre, catechize*
 - *brusquely, docilely, imploringly, beatifically, superciliously*
-  • GCSE texts will stretch **even those who read widely**
- Weaker readers **may not be able to engage** with these texts at all



Many of the new GCSE words are **new roots**

- In popular books, many of the unfamiliar words contain **multiple morphemes**

→ Meanings **can be derived** from the words' constituents: mourn + ful + ly

- In GCSE books, most “new” words are **new roots**

→ Meanings **cannot be derived** from the words' structure

- *aspidistra, crimplene, beseech, coccidia, gambol*



- Pupils **must rely on context or instruction** to understand these words
- Deriving meaning from context requires **advanced language and reading skills**

Is GCSE English Lit **fit for purpose**?

GCSE English Literature should enable students to. . .

- read in depth, critically, and evaluatively
- develop the habit of reading widely and often
- appreciate the depth and power of the English literary heritage

Yet...

- Less than 30% of children read daily in their free time
- Many will lack the skills required to access the GCSE course
- Curriculum review **should consider** the huge **variation in children's language and reading skills...**
- 2024 review: "promote a wider range of literary voices while retaining core texts [...] without adding to the mandatory volume of content"

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- Curriculum review **should consider** the huge **variation in children's language and reading skills...**
- 2024 review: "promote a wider range of literary voices while retaining core texts [...] without adding to the mandatory volume of content"; **no mention of variation in skills**



There is a **virtuous cycle between reading skills and reading motivation**



Original Article



The Children and Young People's Books Lexicon (CYP-LEX): A large-scale lexical database of books read by children and young people in the United Kingdom

Maria Korochkina ¹, Marco Marelli ², Marc Brysbaert ³, and Kathleen Rastle ¹

<https://cyp-lex.rastlelab.com/>



Maria Korochkina & Kathy Rastle
Mar 12, 2024 · 5 min read

What Words do Children Encounter When They Read for Pleasure?

The ability to read opens up worlds. Reading enables children to progress into post-primary education and provides the basis for lifelong...

<https://www.rastlelab.com/blog>



Maria Korochkina & Kathy Rastle
May 19 · 5 min read

What can children learn about morphology from reading for fun?

Article | [Open access](#) | Published: 05 May 2025

Morphology in children's books, and what it means for learning

[Maria Korochkina](#) & [Kathleen Rastle](#)

[npj Science of Learning](#), 10, Article number: 22 (2025) | [Cite this article](#)

4985 Accesses | 23 Altmetric | [Metrics](#)

Psychonomic Bulletin & Review (2025) 33:3
<https://doi.org/10.3758/s13423-025-02830-2>

BRIEF REPORT

Morpheme knowledge is shaped by information available through orthography

Maria Korochkina ¹, Holly Cooper ¹, Marc Brysbaert ², Kathleen Rastle ¹

Journal of Memory and Language 148 (2026) 104746



Contents lists available at [ScienceDirect](#)

Journal of Memory and Language

journal homepage: www.elsevier.com/locate/jml



Morphemes in the wild: Modelling affix learning from the noisy landscape of natural text

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Further reading

THE VOCABULARY
BARRIER IN THE GENERAL
CERTIFICATE OF SECONDARY
EDUCATION (GCSE) IN
ENGLISH LITERATURE

Maria Korochkina¹* and Kathleen Rastle

The English Association Journal
for Teachers of English

THE USE OF english

<https://www.tes.com/magazine>

What makes GCSE English lit so hard for students?

New research has analysed GCSE literature texts against popular fiction – and found three key reasons why so many young people struggle with the qualification



Holly Cooper



Marco Marelli



Marc Brysbaert



Kathy Rastle

Thank you!

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<https://mariakna.github.io/>